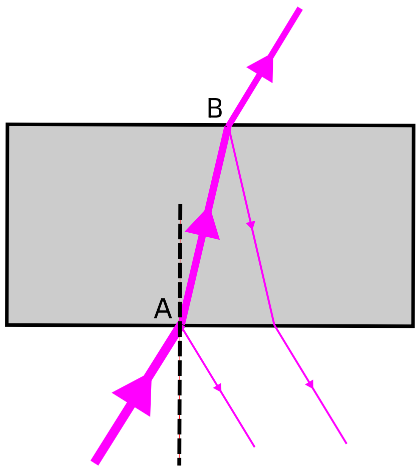
Refraction

When light travels into a ________ material, it changes _____.

This can cause it to change _______.

This is called ________.

1 The diagram below shows a ray of light going through a glass block. Look at the diagram carefully, and then answer the questions.



(a) Draw on the diagram the path the light would take in the glass if it did not change direction at A.

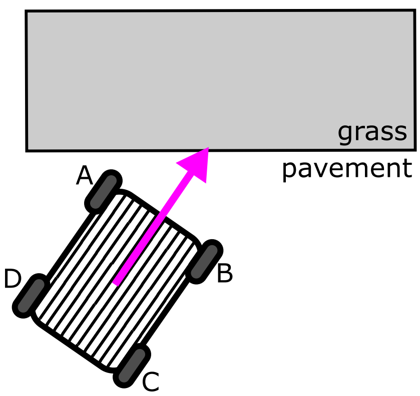
(b) Label the normal with a letter N.

(c) When the light enters the block, which way does it turn? Choose from: **towards the normal** or **away from the normal**.

(d) Describe the direction the light takes as it moves away from the block at B.

(e) Not all light enters the block. What happens to the light which does not enter the block?

2 Refraction is like a trolley being pushed off a pavement onto some grass.



(a) Which wheel on the trolley is going to go onto the grass first? A, B, C or D.

(b) Which way will the trolley turn after the first wheel goes onto the grass and slows down? **Left** or **right**?

(c) What happens after the next wheel goes onto the grass?

(d) Draw an arrow to show the direction of the trolley after two wheels are on the grass.

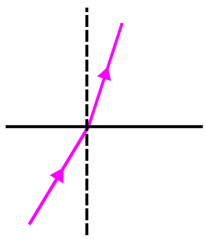
(e) Draw a normal on the diagram.

(f) When the trolley goes onto the grass, which way does it turn? Choose from: **towards the normal** or **away from the normal**.

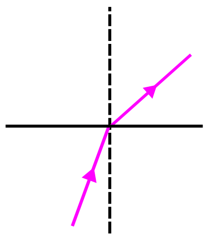
(g) Compare this diagram with the diagram in question 1. Does light slow down or speed up when it enters glass from air?

3 A teacher drew diagrams showing light crossing a boundary from air to glass or glass to air, but forgot to label it! On each diagram, mark with a G the side which is glass.

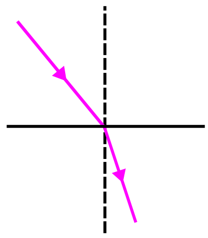
(a)



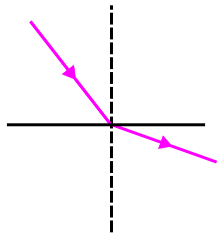
(b)



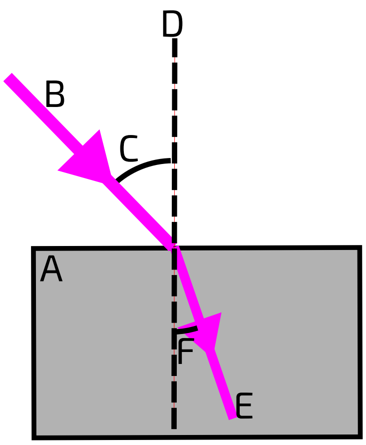
(c)



(d)

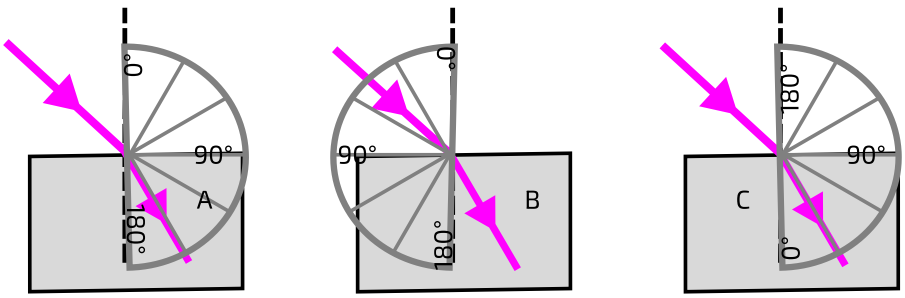


4 Which words describe the features in the diagram below? Fill in the table.



Technical term	**A, B, C, D, E or F?**
glass block	
refracted ray	
incident ray	
angle of refraction	
angle of incidence	
normal	

5 The diagrams below show three attempts to measure an **angle of refraction**.



(a) From which of these protractors could we read off the correct angle?

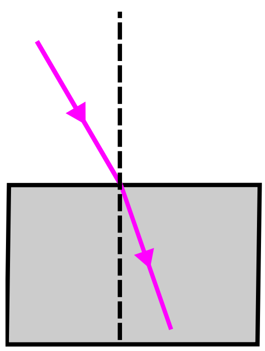
(b) State the angle of refraction of the ray.

(c) Estimate the angle of incidence of the ray. Is it $30°$, $45°$ or $60°$?

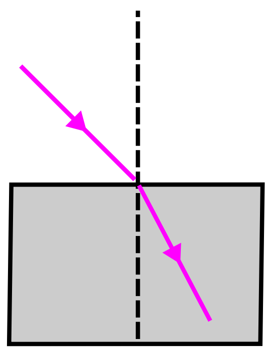
(d) Not all of the light refracts. Draw a reflected ray onto the diagram.

6 In each diagram below, measure and write down the angles of incidence and refraction.

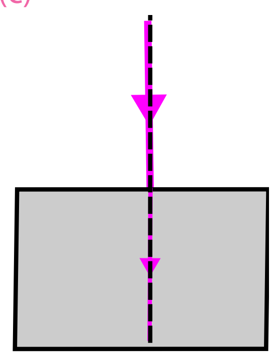
(a)



(b)

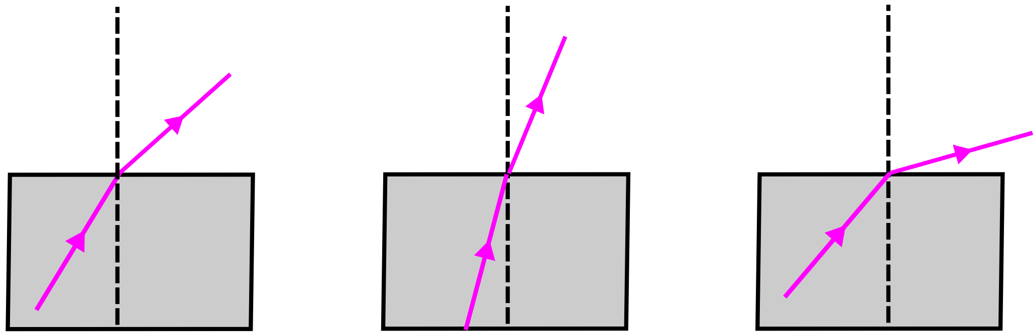


(c)



7 In each diagram below, write down the angles of incidence and refraction. Here the angles of incidence are inside the block of glass.

(a) (b) (c)



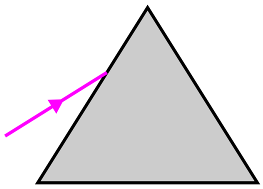
8 Complete the sentences using the phrases **will always**, **does not**, **in to**, **out from**, **incidence**, **refraction**. You do not need to use all of the phrases.

If the angle of incidence is zero, the light ____________ change direction.

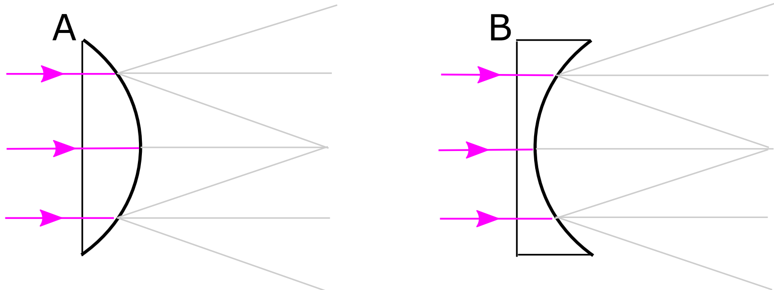
Otherwise, the angle of incidence is larger than the angle of ____________ when light passes __________ a glass block.

The angle of ____________ is smaller than the angle of refraction when light passes __________ a glass block.

9 Draw two further rays on the diagram below to show where the light might go after entering this triangular block (called a _____).



10 The diagrams below show rays passing into pieces of glass with a curved surface.



(a) Draw normals for each of the places where light reaches the curved surface.

(b) Draw the refracted rays to show how light leaves the glass. Each refracted ray follows one of the pale grey lines, but you will need to choose which one to use.

(c) These objects are called _____. A ______ lens is thicker in the middle than at the edge. Which lens is the convex lens? Write A or B.

(d) Complete the sentence: A convex lens causes parallel rays of light to